Math 112: Discrete Mathematics

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1 The Foundations

1.1 Propositions

DEFINITION 1.1. A **proposition** is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

Examples:

- Washington, D.C., is the capital of the United States of America.
- Toronto is the capital of Canada.
- $1 + 1 = 2$.
- $2 + 2 = 3$.
- What time is it?
- Read this carefully.
- $x + 1 = 2$.
- $x + y = z$.

We use letters to denote propositional variables (or sentential variables), that is, variables that represent propositions, just as letters are used to denote numerical variables. The conventional letters used for propositional variables are $p, q, r, s, \ldots$. The truth value of a proposition is true, denoted by T, if it is a true proposition, and the truth value of a proposition is false, denoted by F, if it is a false proposition. Propositions that cannot be expressed in terms of simpler propositions are called atomic propositions.

The area of logic that deals with propositions is called the **propositional calculus** or **propositional logic**. It was first developed systematically by the Greek philosopher Aristotle more than 2300 years ago.

New propositions, called **compound propositions**, are formed from existing propositions using **logical operators**.

1.2 Logical Operations

DEFINITION 1.2 (NEG)**.** Let p be a proposition. The negation of p, denoted by $\neg p$ (also denoted by $\overline{p}$), is the statement "It is not the case that p." The proposition $\neg p$ is read "not p." The truth value of the negation of p, $\neg p$, is the opposite of the truth value of p.

Examples:

- If p= "Michael's PC runs Linux", then $\neg p$= "Michael's PC does not run Linux"

- If q= "Vandana's smartphone has at least 32 GB of memory", then $\neg q$= "Vandana's smartphone has less than 32 GB of memory".

p	$\neg p$
T	F
F	T

DEFINITION 1.3 (AND)**.** Let p and q be propositions. The conjunction of p and q, denoted by $p \wedge q$, is the proposition "p and q." The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

DEFINITION 1.4 (OR)**.** Let p and q be propositions. The disjunction of p and q, denoted by $p \vee q$, is the proposition "p or q." The conjunction $p \vee q$ is true when at least one of p and q is true and is false otherwise.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

DEFINITION 1.5 (XOR)**.** Let p and q be propositions. The exclusive or of p and q, denoted by $p \oplus q$, is the proposition "p exlcusive or q." The conjunction $p \vee q$ is true when exactly one of p and q is true and is false otherwise.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

DEFINITION 1.6 (Conditional)**.** Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition "if p, then q." The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Example:

- Let p be the statement "Maria learns discrete mathematics" and q the statement "Maria will find a good job." Express the statement $p \to q$ as a statement in English.

  Solution: If Maria learns discrete mathematics, then she will find a good job."

- What is the value of the variable x after the statement

  $$\text{if } 2+2=4 \text{ then } x := x+1$$

  if $x = 0$ before this statement is encountered? (The symbol := stands for assignment. The statement $x := x + 1$ means the assignment of the value of $x + 1$ to x.)

DEFINITION 1.7 (converse, contrapositive,inverse)**.** Consider the conditional statement $p \to q$.

- Its **converse** is $q \to p$,
- Its **contrapositive** is $\neg q \to \neg p$,
- Its **inverse** is $\neg p \to \neg q$.

Example:

Find the contrapositive, the converse, and the inverse of the conditional statement "The home team wins whenever it is raining." The statement is the same thing as "If it is rainig, then the home team wins".

Solution:

Converse: "If the home team wins, then it is raining".

Contrapositive: "If the home team does not win, then it is not raining."

Inverse: "If it is not raining, then the home team does not win."

DEFINITION 1.8 (Biconditional)**.** Let p and q be propositions. The biconditional statement $p \leftrightarrow q$ is the proposition "p if and only if q." The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. Biconditional statements are also called bi-implications.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example: Construct the truth table of the compound proposition
$(p \vee \neg q) \rightarrow (p \wedge q)$.

1.3 Precedence

operator	precedence
$\neg$	1
$\wedge, \vee$	2
$\rightarrow$	3
$\leftrightarrow$	4

1.4 Logic and Bit Operations

truth value	bit
T	1
F	0

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
1	1	1	1	0
1	0	1	0	1
0	1	1	0	1
0	0	0	0	0

DEFINITION 1.9. A bit string is a sequence of zero or more bits. The length of this string is the number of bits in the string.

Example. 101010011 is a bit string of length nine.

Example. Find the bitwise OR, bitwise AND, and bitwise XOR of the bit strings 01 1011 0110 and 11 0001 1101.

01 1011 0110 11 0001 1101	
11 1011 1111	bitwise OR
01 0001 0100	bitwise AND
10 1010 1011	bitwise XOR

1.5 Tutology, Contradiction,..

DEFINITION 1.10. A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a **tautology**. A compound proposition that is always false is called a **contradiction**. A compound proposition that is neither a tautology nor a con- tradiction is called a contingency.

Example

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Therefore, for every p, $p \vee \neg p$ is a tautology and $p \wedge \neg p$ is a contradiction.

1.6 Logical Equivalence

DEFINITION 1.11. The compound propositions p and q are called logically equivalent if $p \leftrightarrow q$ is a tautology. The notation $p \equiv q$ denotes that p and q are logically equivalent.

Example: (De Morgan's Laws)

- $\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$
- $\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$

Example: $p \rightarrow q$ and $(\neg p) \vee q$ are logically equivalent.

Example: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$. Similarly, we have $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

Example: Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

Example: Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

1.7 Satsifiability

DEFINITION 1.12. A compound proposition is **satisfiable** if there is an assignment of truth values to its variables that makes it true. When no such assignments exists, that is, when the compound proposition is false for all assignments of truth values to its variables, the compound proposition is **unsatisfiable**.

When we find a particular assignment of truth values that makes a compound proposition true, we have shown that it is satisfiable; such an assignment is called a solution of this particular satisfiability problem. However, to show that a compound proposition is unsatisfiable, we need to show that every assignment of truth values to its variables makes it false.

Example: Determine whether each of the compound propositions $(p \vee \neg q) \wedge (q \wedge \neg r) \wedge (r \vee \neg p)$, $(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$, and $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$ is satisfiable.

1.8 Quantifiers

DEFINITION 1.13. The universal quantification of $P(x)$ is the statement

"$P(x)$ for all values of x in the domain."

We write $\forall x P(x)$. $\forall$ is called the universal quantifier. If x is an element such that $P(x)$ is false we say that x is a counterexample to $\forall x P(x)$.

DEFINITION 1.14. The existential quantification of $P(x)$ is the statement
"There exists an element x in the domain such that $P(x)$."
We write $\exists x P(x)$. $\exists$ is called the existential quantifier.

Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for all x	$P(x)$ is false for some x.
$\exists x P(x)$	There is an x : $P(x)$ is true	$P(x)$ is false for every x

Example: Let $P(x)$ be the statement "$x+1 > x$". If the domain consists of all real numbers, then $\forall x P(x)$ is true.

Example: Let $Q(x)$ be the statement $x < 2$. If the domain is $\mathbb{R}$, then $\forall x Q(X)$ is false.

Example: Let $R(x)$ be the statement $x^2 > 0$. Then, if the domain is $\mathbb{R}$, then $\forall x R(x)$ is false. 0 is a counterexample to the statement $\forall x R(x)$.

The truth value may depend on the domain!

Example: What is the truth value of $\forall x(x^2 \geq x)$ if the domain consists of all real numbers? What is the truth value of this statement if the domain consists of all integers?

When the domain is $\mathbb{R}$, .5 is a counterexample to $\forall x P(x)$. Hence $\forall x P(x)$ is false. When the domain is $\mathbb{Z}$ then $\forall x P(x)$ is true. In order to emphasize the domain sometimes I write $\forall x \in \mathbb{Z}\ \ P(x)$.

Example: Let $P(x)$ be the statement that $x > 3$. Then $\exists x P(x)$ is true if the domain is $\mathbb{R}$.

Example: Let $Q(x)$ be the statemnt $x = x+1$ (the domain is $\mathbb{R}$). Then $\exists x Q(x)$ is false.

Uniqueness quantifier: $\exists!$.

$\exists! x P(x)$ means there is one and only one x such that $P(x)$.

Example: Let $P(x)$ be the statement $x^3 = 8$ (the domain is $\mathbb{R}$). Then $\exists! x P(x)$ is true.

Example: Let $Q(x)$ be the statement $x^2 = 1$ (the domain is $\mathbb{R}$). Then $\exists! x Q(x)$ is false. Because $-1 \neq 1$ and $(-1)^2 = 1^2 = 1$.

DEFINITION 1.15. Statements involving predicates and quantifiers are logically equivalent if and only if they have the same truth value no matter which predicates are substituted into these statements and which domain of discourse is used for the variables in these propositional functions. We use the notation $S \equiv T$ to indicate that two statements S and T involving predicates and quantifiers are logically equivalent.

Example: The statements $\forall x(P(x) \wedge Q(x))$ and $\forall x P(x) \wedge \forall x Q(x)$ are logically equivalent...

Fact: $\neg(\forall x P(x)) \equiv \exists x \neg P(x)$, and
$\neg(\exists x P(x)) \equiv \forall x \neg P(x)$.

Example: The negation of $\forall x(x^2 > x)$ is $\exists x(x^2 \leq x)$.

The negation of $\exists x(x^2 = 2)$ is $\forall x(x^2 \neq 2)$.

Example: $\neg\forall x(P(x) \to Q(x))$ is logically equivalent to $\exists x(P(x) \wedge \neg Q(x))$.

1.9 Nested Quantifiers

Examples:

- $\forall x\exists y(x + y = 0)$.
- $\forall x\forall y(x + y = y + x)$.
- $\forall x\forall y((x > 0) \wedge (y < 0) \to (xy < 0))$.
- $\forall x\forall y P(x, y) \equiv \forall y\forall x P(x, y)$! But, in general, the order of quantifiers is important!
- $\forall x\exists y(x + y = 0)$ is true for every value of x and y in the domain $\mathbb{R}$. But $\exists x\forall y(x + y = 0)$ is false.
- Let $Q(x, y, z)$ be the statement $x+y = z$ (domain is $\mathbb{R}$). Then $\forall x\forall y\exists z Q(x, y, z)$ is true and $\exists z\forall x\forall y Q(x, y, z)$ is false. Why?
- The statement "the sum of two positive integers is always positive" can be expressed as $\forall x\forall y((x > 0) \wedge (y > 0)) \to (x + y > 0)$. If we restrict the domain to the set of positive numbers, $\mathbb{R}_{>0}$, then it can be expressed simply by $\forall x\forall y(x + y > 0)$.
- In calculus, $\lim_{x\to a} f(x) = A$ means
$$\forall\varepsilon > 0\exists\delta > 0\forall x(0 < |x - a| < \delta) \to (|f(x) - A| < \varepsilon).$$
Similarly, f is continuous at a if
$$\forall\varepsilon > 0\exists\delta > 0\forall x(|x - a| < \delta) \to (|f(x) - f(a)| < \varepsilon).$$

Example: Express the negation of the statement $\forall x\exists y(xy = 1)$ so that no negation precedes a quantifier.

$(\exists x\forall y)(xy \neq 1))$

Example Use quantifiers to express the statement that "There does not exist a woman who has taken a flight on every airline in the world." $(\neg\exists w\forall a(P(w, f) \wedge Q(f, a))$, where $P(w, f)$ is "w has taken f" and $Q(f, a)$ is "f is a flight on a".)

Example Use quantifiers and predicates to express the fact that $\lim_{x\to a} f(x)$ does not exist where $f(x)$ is a real-valued function of a real variable x and a belongs to the domain of f.

$$\forall L\exists\varepsilon > 0\forall\delta > 0\exists x[(0 < |x - a| < \delta) \wedge (|f(x) - L| \geq \varepsilon)].$$

More Examples:

- $C(x)$ is "x has a computer" and $F(x, y)$ is "x and y are friends" the domain consists of students in the school. Then
$$\forall x(C(x) \vee \exists y(C(y) \wedge F(x, y)))$$
can be translated into English as "every student in the school has a computer or has a friend with a computer".

- The domain is again the set of students in the school $F(x, y)$ is as above. What is the meaning of
$$\exists x \forall y \forall z((F(x, y) \wedge F(x, z) \wedge (y \neq z) \rightarrow \neg F(y, z))?$$

1.10 Rules of Inference

DEFINITION 1.16. An **argument** in propositional logic is a sequence of propositions. All but the final proposition in the argument are called **premises** and the final proposition is called the **conclusion**. An argument is valid if the truth of all its premises implies that the conclusion is true. An **argument form** in propositional logic is a sequence of compound propositions involving propositional variables. An argument form is **valid** if no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.

From the definition of a valid argument form we see that the argument form with premises $p_1, p_2, \cdots, p_n$ and conclusion q is valid exactly when $(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \rightarrow q$ is a tautology.

The tautology $(p \wedge (p \rightarrow q)) \rightarrow q$ is the basis of the rule of inference called **modus ponens**, or the **law of detachment**.

$$\begin{array}{lll} p & & \\ p & \rightarrow & q \\ \hline \therefore & q & \end{array}$$

Example: " If $\sqrt{2} > \frac{3}{2}$, then $(\sqrt{2})^2 > \left(\frac{3}{2}\right)^2$". We know that $\sqrt{2} > \frac{3}{2}$. Hence we have $2 > \frac{9}{4}$.

The argument form is valid. But, one of the premises is false. Therefore we cannot conclude that the conlusion is true! Note that the conlusion is also false.

Rule of inference	Tautology	Name
p $p \to q$ $\therefore\ q$	$(p \wedge (p \to q)) \to q$	Modus ponens
$\neg q$ $p \to q$ $\therefore\ \neg p$	$(\neg q \wedge (p \to q)) \to \neg p$	Modus tollens
$p \to q$ $q \to r$ $\therefore\ p \to r$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical syllogism
$p \vee q$ $\neg p$ $\therefore\ q$	$((p \vee q) \wedge \neg p) \to q$	Disjunctive syllogism
p $\therefore\ p \vee q$	$p \to (p \vee q)$	Addition
$p \wedge q$ $\therefore\ p$	$(p \wedge q) \to p$	Simplification
p q $\therefore\ p \wedge q$	$((p) \wedge (q)) \to (p \to q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore\ q \wedge r$	$((p \vee q) \wedge (\neg p \vee r)) \to (q \vee r)$	Resolution

Example: State which rule of inference is the basis of the following argument: "It is below freezing and raining now. Therefore, it is below freezing now."

Example: State which rule of inference is used in the argument: If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow. Therefore, if it rains today, then we will have a barbecue tomorrow.

Example: Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."

Examples: Show that the premises "If you send me an e-mail message, then I will finish writing the program," "If you do not send me an e-mail message, then I will go to sleep early," and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed."

Computer programs have been developed to automate the task of reasoning and proving theorems. Many of these programs make use of a rule of inference known

as **resolution**. This rule of inference is based on the tautology

$$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r).$$

The final disjunction in the resolution rule, $q \vee r$, is called the **resolvent**.

Example: Use resolution to show that the hypotheses "Jasmine is skiing or it is not snowing" and "It is snowing or Bart is playing hockey" imply that "Jasmine is skiing or Bart is playing hockey."

Remark: To construct proofs in propositional logic using resolution as the only rule of inference, the hypotheses and the conclusion must be expressed as **clauses**, where a clause is a disjunction of variables or negations of these variables. We can replace a statement in propositional logic that is not a clause by one or more equivalent statements that are clauses.

Example: Suppose we have a statement of the form $p \vee (q \wedge r)$. Because $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$, we can replace the single statement $p \vee (q \wedge r)$ by two statements $p \vee q$ and $p \vee r$, each of which is a clause.

We can also replace a statement of the form $\neg(p \vee q)$ by the two statements $\neg p$ and $\neg q$. (DeMorgan)

We can also replace a conditional statement $p \rightarrow q$ with the equivalent disjunction $\neg p \vee q$.

Example: Show that the premises $(p \wedge q) \vee r$ and $r \rightarrow s$ imply the conclusion $p \vee s$.

1.11 Fallacies

1.11.1

The proposition $((p \rightarrow q) \wedge q) \rightarrow p$ is not a tautology, because it is false when p is false and q is true. However, there are many incorrect arguments that treat this as a tautology. In other words, they treat the argument with premises $p \rightarrow q$ and q and conclusion p as a valid argument form, which it is not. This type of incorrect reasoning is called the **fallacy of affirming the conclusion.**

Example: Is the following argument valid?

If you do every problem in this book, then you will learn discrete mathematics. You learned discrete mathematics.

Therefore, you did every problem in this book.

1.11.2

The proposition $((p \rightarrow q) \wedge \neg p) \rightarrow \neg q$ is not a tautology, because it is false when p is false and q is true. Many incorrect arguments use this incorrectly as a rule of inference. This type of incorrect reasoning is called the **fallacy of denying the hypothesis.**

Example:

Let p and q be as in the previous example. If the conditional statement $p \to q$ is true, and $\neg p$ is true, is it correct to conclude that $\neg q$ is true? In other words, is it correct to assume that you did not learn discrete mathematics if you did not do every problem in the book, assuming that if you do every problem in this book, then you will learn discrete mathematics?

1.11.3

$\forall x P(x), \therefore P(c)$	Universal instantiation
$P(c)$ for any c, $\therefore \forall x P(x)$	Universal generalization
$\exists x P(x)$, $\therefore P(c)$ for some element c	Existential instantiation
$P(c)$ for some element c, $\therefore \exists x P(x)$	Existential generalization

Example: Show that the premises "Everyone in this discrete mathematics class has taken a course in computer science" and "Marla is a student in this class" imply the conclusion "Marla has taken a course in computer science."

Example: Show that the premises "A student in this class has not read the book," and "Everyone in this class passed the first exam" imply the conclusion "Someone who passed the first exam has not read the book."

Because universal instantiation and modus ponens are used so often together, this combination of rules is sometimes called **universal modus ponens**.

$$\forall x(P(x) \to Q(x))$$
$$P(a) \text{ where } a \text{ is a particular element}$$
$$\therefore Q(a)$$

Example:

Assume that "For all positive integers n, if n is greater than 4, then n^2 is less than 2^n" is true. Use universal modus ponens to show that $100^2 < 2^100$.

2 Sets, Functions, etc

2.1 The set $\mathbb{N}$

The set $\mathbb{N} = \{0, 1, 2, ...\}$ is called the set of natural numbers. Everybody knows what they are. Here we want to emphesize some of the less well-know facts about $\mathbb{N}$.

FACT 2.1 (Well-Ordering Principle)**.** If A is a non-empty subset of $\mathbb{N}$, then A has a least element.

PROPOSITION 2.2 (Mathematical Induction)**.** *If $A \subseteq \mathbb{N}$ and if*

1. $0 \in A$,
2. $n \in A \implies (n+1) \in A$,

then $A = \mathbb{N}$.

Proof. Suppose that A satisfies the properties above but $A \neq \mathbb{N}$. Let $B = \mathbb{N} - A = \{n \in \mathbb{N} : n \notin A\}$. Then B is a non-empty subset of $\mathbb{N}$. By the Well-Ordering Principle, B has a least element, say b. We have two cases:

1. $b = 0$. This is not possible because A has 0 according to the hypothesis.
2. $b \neq 0$. Then $b-1 \in A$. But, according to the hypothesis, $b = (b-1)+1 \in A$, which also leads to a contradiction.

Therefore, $A = \mathbb{N}$. □

2.2 Sets

By a set we mean a collection of objects. Sometimes we write sets as a list

$$A = \{1, 2, 5, 7\},$$

sometimes with a defining property

$$B = \{n \in \mathbb{Z} : n \text{ is odd}\}$$

or

$$C = \{(x, x^2) : x \in \mathbb{R}\}.$$

If x is a member of a set A, we write $x \in A$. Otherwise we write $x \notin A$. The set without any element is called the empty set and is denoted by $\{\}$ or $\emptyset$. If A and B are sets and if every member of A is a member of B, we say that A is a subset of B and we write $A \subseteq B$. In general, in this course, we may think that our sets are subsets of a large enough set U. We have

$$(A \subseteq B) \Longleftrightarrow (\forall x)(x \in A \implies x \in B).$$

We define $A = B$ if $A \subseteq B$ and $B \subseteq A$. Therefore,

$$A = B \Longleftrightarrow (A \subseteq B \wedge B \subseteq A).$$

If A and B are sets, we define $A \cap B$ and $A \cup B$ by

$$(\forall x) x \in A \cap B \Longleftrightarrow (a \in A \wedge x \in B)$$

and

$$(\forall x) x \in A \cup B \Longleftrightarrow (x \in A \vee x \in B).$$

We also define

$$A^c = \{x : x \notin A\}$$

and

$$A - B = \{x : x \in A \text{ and } x \notin B\}.$$

We say that A and B are disjoint if $A \cap B = \emptyset$. In this case $A \cup B$ is called a disjoint union.

If X and Y are two sets, we form a new set $X \times Y$ as follows:

$$X \times Y = \{(x, y) : x \in X \wedge y \in Y\}.$$

By a relation R from X to Y we mean a subset of $X \times Y$.

Example 2.3. Let $R = \{(x, y) \in \mathbb{R}^2 = \mathbb{R} \times \mathbb{R} : x \leq y\}$. Then R is a relation from $\mathbb{R}$ to $\mathbb{R}$. It should be trivial that $(x, y) \in R$ iff $x \leq y$.

Example 2.4. Consider $C = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$. Then C is the unit circle in $\mathbb{R}^2$. $(x, y) \in C$ iff (x, y) are the coordinates of some point on the circle.

2.3 Functions

DEFINITION 2.5. A relation f from X to Y is a function if

1. $(\forall x \in X)(\exists y \in Y)$ such that $(x, y) \in f$, and
2. if $(x, y) \in f$ and $(x, z) \in f$ then $y = z$.

Therefore, f is a function iff $(\forall x \in X)(\exists! y \in Y)(x, y) \in f$.
If f is a function we write $f : X \to Y$ is a function and $y = f(x)$ for $(x, y) \in f$.

Example 2.6.

1. C (the unit circle above) is not a function. (Why?)
2. $\{(x, 3x) : x \in \mathbb{R}\}$ is a function.
3. $\{(x, x^2) : x \in \mathbb{R}\}$ is a function.

4. Let X be any set, $A \subseteq X$. Then $1_A : X \to \{0, 1\}$ defined by

$$1_A(x) = 1 \text{ iff } x \in A$$

is a function. It is called the characteristic function of A.

DEFINITION 2.7. Let X and Y be sets, $F : X \to Y$ be a function.

1. f is said to be 1-1 (injective) if $x \neq y \implies f(x) \neq f(y)$. In other words, f is 1-1 if distinct elements of X go under f to distinct elements in Y.

2. f is said to be onto (surjective) if for every $y \in Y$ there exists some $x \in X$ such that $f(x) = y$.

Remark 2.8. By using the equivalence $(p \Rightarrow q) \equiv (\neg q \Rightarrow \neg p)$, we see that a function f is 1-1 iff $x = y$ whenever $f(x) = f(y)$.

Example 2.9. 1. $f : \mathbb{R} \to \mathbb{R}$ $f(x) = 3x - 1$ is a 1-1 and onto function. Such functions are said to be bijective or bijection. Sometimes such a function is said to be 1-1 correspondence.

2. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ is a function which is not 1-1 or onto. $f : \mathbb{R} \to [0, \infty)$, $f(x) = x^2$ is onto but not 1-1. The function $f : [0, \infty) \to [0, \infty)$ given by $f(x) = x^2$ is a bijection. Similarly the function $f : [2, \infty) \to [4, \infty)$ isalso a bijection.

Let $f : X \to Y$ be function. If $A \subseteq X$, we put

$$f(A) = \{f(x) : x \in A\}.$$

It is called the image of A under f. SImilarly, if $B \subseteq Y$, we put

$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

It is called the preimage or inverse image of B under f.

If $f : X \to y$ and $g : Y \to Z$ are functions, the funcetion defined by

$$(g \circ f)(x) = g(f(x))$$

is called the composition of f and g.

Exercise 2.10. 1. Show that $(f \circ g) \circ h = f \circ (g \circ h)$. ($f, g, h$ are appropriate function for which the sentence is meaningful.)

2. Prove or disprove: If $g \circ f$ is onto, then g is onto.

3. Prove or disprove: If $g \circ f$ is 1-1, then f is 1-1.

4. Prove or disprove: If f and g are 1-1, then $g \circ f$ is 1-1.

5. Prove or disprove: If f and g are onto, then $g \circ f$ is onto.

6. Prove or disprove: If $g \circ f$ is onto, f is onto.

7. Prove or disprove: If $g \circ f$ is 1-1, then g is also 1-1.

If $f : X \to Y$ and $g : Y \to X$ are two functions such that

$$(\forall x \in X) g(f(x)) = x \text{ and } (\forall y \in Y) f(g(y)) = y,$$

we say that g is the inverse function of f and we write $g = f^{-1}$.

Exercise 2.11. Show that $f : X \to Y$ has an inverse iff it is a bijection.

2.4 Cardinality, the Cantor-Schröder-Bernstein Theorem

We say that two sets X and Y has the same cardinality if there exists a bijection $f : X \to Y$. We write $|X| = |Y|$. If there exists a 1-1 function $f : X \to Y$, we write $|X| \leq |Y|$. (In some sense Y contains a copy of X.

Clearly if $X \subseteq Y$, we have $|X| \leq |Y|$.

THEOREM 2.12 (Cantor-Schröder-Bernstein)**.** *If* $|X| \leq |Y|$ *and* $|Y| \leq |X|$ *then* $|X| = |Y|$.

PROPOSITION 2.13. $|\mathbb{N}| = |\mathbb{Z}|$

Proof. Clearly $|\mathbb{N}| \leq |\mathbb{Z}|$. Define $f : \mathbb{Z} \to \mathbb{N}$ by

$$f(n) = \begin{cases} 1 & \text{if } n = 0 \\ 3^n & \text{if } n > 0 \\ 5^{-n} & \text{if } n < 0. \end{cases}$$

Except 1, powers of 3 and 5 are all distinct. Therefore, f is a 1-1 function from $\mathbb{Z}$ to $\mathbb{N}$. Therefore we also have $|\mathbb{Z}| \leq |\mathbb{N}|$. The result follows by the Cantor-Schröder-Bernstein Theorem. □

If $n \geq 1$ is an integer, we define $[n] = \{1, 2, 3, ..., n\}$. We say that a set X has n elements if $|X| = |[n]|$. We say that a set is finite if it has n elements for some $n \in \mathbb{N}$. Otherwise we say that X is infinite.

Exercise 2.14. 1. Show that the union of two finite sets is finite.

2. Suppose that A and B are sets with A is infinite. Then $A \cup B$ is infinite.

3. If X is infinite and $A \subseteq X$ is finite, then $X - A$ is infinite.

4. Any subset of a finite set is finite.

5. If a set contains an infinite set as a subset, then it is also infinite.

We say that a set X is countable if it is finite or $|X| = |\mathbb{N}|$. Therefore $\mathbb{Z}$ and $\mathbb{Q}$ are countable sets.

Remark 2.15. 1. An infinite set X is countable iff it can be written as $X = \{x_1, x_2, .x_3, ..., x_n, ...\} = \{x_n : n \in \mathbb{N}\}$.
2. A set X is countable iff there is a 1-1 function $\varphi : X \to \mathbb{N}$.

PROPOSITION 2.16. *Any subset of a countable set is countable.*

Proof. □

PROPOSITION 2.17. *If X and Y are countable disjoint sets then $X \cup Y$ is countable.*

Proof. □

Exercise 2.18. Show that $X \cup Y = X \cup (Y - X)$. Deduce that if X and Y are countable (not necessarily disjoint) then $X \cup Y$ is countable.

PROPOSITION 2.19. $|\mathbb{Q}| = |\mathbb{N}|$.

Proof. Let $P = \{x \in \mathbb{Q} : x > 0\}$. Let $-P = \{-x : x \in P\}$. Then $\mathbb{Q}$ is the dijoint union of $P, -P$ and $\{0\}$. Let us show that P is countable. Consider $P = \{\frac{m}{n} : m, n \geq 1, \frac{m}{n} \text{ is reduced}\}$. Define $\varphi : P \to \mathbb{N}$ by $\varphi(\frac{m}{n}) = 3^m 5^n$. Let us see that φ is 1-1. Suppose $\varphi(\frac{m}{n}) = \varphi(\frac{m'}{n'})$. This means that $3^m 5^n = 3^{m'} 5^{n'}$. It follows that $3^{m-m'} = 5^{n'-n}$ Since 3 and 5 have only common power, namely 1, we see that $m = m'$ and $n = n'$. Therefore we have shown that $\varphi(x) = \varphi(y)$ implies that $x = y$. Therefore φ is 1-1 and P is countable. It follows that $P \cup -p$ is also countable. It follows that $\mathbb{Q}$ is countable. □

Example 2.20. Let S be the set of all 0-1 sequences. In other words, S is the set of all functions $x : \mathbb{N} \to \{0, 1\}$. Clearly S is infinite. Let us see that S is not countable.

We prove this stament by contradiction. Suppose that S is countable, therefore we can write $S = \{x_n : n \geq 1, n \in \mathbb{N}\}$. Consider the sequence $x : \mathbb{N} \to \{0, 1\}$ defined by the following: for each $n \in \mathbb{N}$, $x(n) = 1 - x_n(n)$. Therefore, $x \neq x_n$ for every n. Because the nth terms of x and x_n are diffferent. Clearly $x : \mathbb{N} \to \{0, 1\}$ is a sequence, hence $x \in S$, but it is not on the list. This is a contradiction. Therefore our assumption is false. This shows that S is not countable.

Example 2.21. Let X be any set. Let $\mathcal{P}(X)$ be the set of all subsets of X. It is called the power set of X. Let $2^X = \{0, 1\}^X = \{f : X \to \{0, 1\}\}$. We have a function $F : \mathcal{P}(X) \to 2^X$ given by $F(A) = 1_A$ (the characteristic function of A.) We claim that F is a bijection.

1. F is 1-1: Suppose that $F(A) = F(B)$. Then $F(A)(x) = F(B)(x)$ for all $x \in X$. This shows that $F(A)(x) = 1$ iff $F(B)(x) = 1$. In other wards, $x \in A$ iff $x \in B$. This says that $A = B$. Therefore, we proved that $F(A) = F(B)$ implies $A = B$. Hence F is 1-1.

2. F is onto: Let $g \in 2^X$. Then let $A = \{x \in X : g(x) = 1\}$. Then $x \in A$ iff $g(x) = 1$. This shows that $g = F(A)$. Therefore we proved that every $g \in 2^X$ is the image of some element in $\mathcal{P}(X)$. Hence F is onto.

Remark 2.22. It follows that the set S in an earlier example (S is the set of all 0-1 sequences) and $\mathcal{P}(\mathbb{N})$ have the same cardinality. Since S is not countable, $\mathcal{P}(\mathbb{N})$ is not countable. It is an easy exercise to show that if $|X| = |Y|$ and if X is countable, then Y is also countable. THEREFORE, WE SEE THAT THERE ARE INFINITE SETS WITH DIFFERENT CARDINALITIES.

PROPOSITION 2.23. *For any set X, $|X| \neq |\mathcal{P}(X)|$.*

Proof. We prove by contradiction. Suppose that there exists a bijection $\varphi : X \to \mathcal{P}(X)$. Consider the subset $A = \{x \in X : x \notin \varphi(x)\}$. Since φ is onto, there is some $a \in X$ such that $A = \varphi(a)$. Then we have

$$a \in A \Longleftrightarrow a \notin A.$$

Clearly this is a contradiction. Therefore, there is no bijection between X and $\mathcal{P}(X)$. □

Conclusion 2.24. $\mathbb{N}$, $\mathcal{P}(\mathbb{N})$, $\mathcal{P}(\mathcal{P}(\mathbb{N}))$,... all have different cardinalities. THEREFORE THERE ARE INFINTELY MANY DIFFERENT INFINITE CARDINALITIES.

CONJECTURE 2.25 (Continuum Hypothesis)**.** There is no cardinality between $|\mathbb{N}|$ and $|\mathcal{P}(\mathbb{N})|$.

2.5 Proof by Induction

The Principle of Induction can be stated in same equivalent ways. One of the convenient form is the following:

FACT 2.26 (Principle of Induction)**.** Suppse that A is a set of integers such that

1. $1 \in A$
2. $n \in A \implies (n+1) \in A$.

Then $n \in A$ for every integer $n \geq 1$.

Actually, one can replace 1 by any integer:

FACT 2.27 (Principle of Induction)**.** Let k be an integer. Suppse that A is a set of integers such that

1. $k \in A$
2. $n \in A \implies (n+1) \in A$.

Then $n \in A$ for every integer $n \geq k$.

2.6 Strong Induction

Suppose that for every integer $n \geq 1$, $P(n)$ is a mathematical statement, and that the following two conditions are satisfied:

1. $P(1)$ is true.
2. For every integer $k \geq 1$, the statements $P(1), P(2), ..., P(k)$ imply $P(k+1)$.

Then $P(n)$ is true for all $n \geq 1$.

2.7 Examples

Example 2.28. For $n \in \mathbb{N}$ we have

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Let A be the set of all $n \in \mathbb{N}$ such that the equation above is true. We want to show that $A = \mathbb{N}$.

1. $n = 1$: This is trivial.
2. Suppose that $\sum_{i=1}^{n} i = n(n+1)/2$. Then we have

$$\sum_{i=1}^{n+1} i = \left(\sum_{i=1}^{n} i \right) + (n+1) = (n(n+1)/2) + (n+1) = (n+1)(n+2)/2.$$

The proof is complete.

□

Example 2.29. If $n \geq 1$, we have

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1.$$

Proof. 1. $n = 1$: This case is trivial.

2. Suppose that the equality above is true for $n \in \mathbb{N}$. Then we have

$$\sum_{i=0}^{n+1} 2^i = \left(\sum_{i=0}^{n} 2^i \right) + 2^{n+1} = 2^{n+2} - 1.$$

□

Example 2.30. If $r \neq 1$ and $n \in \mathbb{N}$, we have

$$1 + r + r^2 + \cdots + r^n = \frac{r^{n+1} - 1}{r - 1}.$$

Proof. 1. The case $n = 1$ is trivial.

2. Suppose we have the equality for n. Then, we have

$$1 + r + \cdots + r^{n+1} = \frac{r^{n+1} - 1}{r - 1} + r^{n+1} = \frac{r^{n+2} - 1}{r - 1}.$$

□

Example 2.31. 1. For $n \geq 1$ we have $n < 2^n$.

2. For $n \geq 4$ we have $2^n < n!$

3. For $j \geq 1$ put

$$H_j = 1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{j}.$$

We claim that

$$H_{2^n} \geq 1 + \frac{n}{2}$$

whenever $n \geq 1$.

4. For $n \geq 1$ we claim that $n^3 - n$ is divisible by 3.

Example 2.32. Show that $7^{n+2} + 8^{2n+1}$ is always divisible by 57 whenever n is a nonnegative integer.

Proof. 1. $n = 0$ is trivial.

2. Suppose that $7^{n+2} + 8^{2n+1} = 57K$ for sme integer K. Then,

$$7^{n+3} + 8^{2(n+1)+1} = 7.7^{n+2} + 64.8^{2n+1} = 57(7K + 8^{2n+1}).$$

□

Example 2.33. If a set S has n elements, then its power set $\mathcal{P}(S)$ has 2^n elements.

Proof. 1. $P(0)$ is true.

2. Suppose that $P(n)$ is true. Let T be a set with $n + 1$ elements, let $x \in T$ and $S = T \backslash \{x\}$. Then, we have $\mathcal{P}(T) = \mathcal{P}(S) + \{A \cup \{x\} : A \in \mathcal{P}(S)\}$. Therefore $\mathcal{P}(T)$ has $2.2^n = 2^{n+1}$ elements.

□

FACT 2.34 (Strong Induction)**.** Suppose that

1. $P(0)$ is true.

2. For every positive k the conditional statement

$$P(1) \wedge P(2) \wedge ... \wedge P(k) \rightarrow P(k + 1)$$

is true.

Then, the statement $P(n)$ is true for every nonnegative integer.

Example 2.35. If n is an integer greater than 1, then n can be written as a product of primes.

Proof. 1. $P(2)$ is true.

2. Suppose that $P(2), ...P(n)$ are all true. If $n+1$ is prime, there is nothing to prove! If $n+1$ is not a prime, then $n+1 = ab$ where a and b are integers greater than 1 and smaller than $n+1$. By hypthesis both a and b are products of primes. The result follows.

□

Example 2.36. Prove that every amount of postage of 12 cents or more can be formed using just 4-cent and 5-cent stamps.

The proof will be by strong induction on n. The case $n = 12 = 3 \cdot 4$ is easy. The cases $13 = 2 \cdot 4 + 1 \cdot 5$, $14 = 1 \cdot 4 + 2 \cdot 5$, $15 = 3 \cdot 5$, $16 = 4 \cdot 4$ are also easy. Suppose that we know that $n \geq 16$ and the statemnt is true for $k = 16, 17, ..., n$. Then the result is true for $n - 4$. This means that there are natural numbers x and y such that $n - 1 = x \cdot 4 + y \cdot 5$. But, the, it follows that $n = (x+1) \cdot 4 + y \cdot 5$. This completes the proof.

3 Basic Counting Methods

3.1 Product Rule

FACT 3.1 (Product Rule)**.** *Suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and for each of these ways of doing the first task, there are n_2 ways to do the second task, then there are $n_1 n_2$ ways to do the procedure.*

Example 3.2. A new company with just two employees, Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Example 3.3. There are 32 computers in a data center in the cloud. Each of these computers has 24 ports. How many different computer ports are there in this data center?

Example 3.4. How many different bit strings of length seven are there?

Example 3.5. How many functions are there from a set with m elements to a set with n elements?

Example 3.6. How many 1-1 functions are there from a set with m elements to a set with n elements?

Solution: The answer is $n \cdot (n-1) \cdot (n-1) \cdots (n-m+1) = \frac{n!}{(n-m)!}$. We remark that if $n < m$ some of these terms will be 0, hence there will be no 1-1 function from A to B. □

Example 3.7. Use the product rule to show that the number of different subsets of a finite set S is $2^{|S|}$.

Solution: Let $|S| = n$. Thus, S is a finite set. We know that there is a 1-1 correspondence between the power set of S and the set of functions from S to $\{0, 1\}$. It should be clear that there is a 1-1 correspondence between thhe set of functions from S to $\{0, 1\}$ and the set of bit strings of length $|S|$. Now we know from a previous example that the number of bit strings of length $|S$ is $2^{|S|}$. Therefore,

$$|2^S| = 2^{|S|}.$$

□

Example 3.8. Chalk comes in three different lengths, eight different colors, and four different diameters. How many different kinds of chalk are there?

Solution: The answer is

$$3 \times 8 \times 4 = 96.$$

□

Example 3.9. Determine the number of positive integers that divides the number

$$2^4 \times 3^3 \times 5^4 \times 11^2.$$

Solution: Let $N = 2^4 \times 3^3 \times 5^4 \times 11^2$. If n is a positive integer that divides N, then it has only prime divisors from the list $\{2, 3, 5, 11\}$. Moreover, it should be of the form

$$n = 2^a \times 3^b \times 5^c \times 11^d$$

where

$$0 \leq a \leq 4; \quad 0 \leq b \leq 3, \quad 0 \leq c \leq 4 \text{ and } 0 \leq d \leq 2.$$

Therefore we have 5 choices for a, 4 choices for b, 5 choices for c and 3 choices for d. Hence, we have $5 \times 4 \times 5 \times 5 \times 3 = 300$ choices.

□

Example 3.10. How many two digit numbers have distinct and nonzero digits?

Solution: Any two digit number is of the form ab where a and b are integers in $\{0, 1, ..., 9\}$. There are 9 possibilities for a non-zero digit. We want them to be distinct. Therefore, we have

$$9 \times 8 = 72$$

possibilities.

There is another method: The number of two digit numbers $\{10, 11, ..., 99\}$ is 90. Nine of them have zeros: 10,20,...,90. Also nine of them have identical digits: 11,22,33,...,99. Observe that they don't have a common element. Therefore, the number of two digit numbers with different and nonzero digits is

$$90 - 9 - 9 = 72.$$

□

FACT 3.11 (Product Rule; General Form)**.**

$$|A_1 \times A_2 \times \cdots A_m| = |A_1| \cdot |A_2| \cdots |A_m|.$$

3.2 Sum Rule

FACT 3.12 (Sum Rule)**.** *Suppose that a set is partitioned into pairwise disjoint parts $S_1, S_2, ..., S_m$.The number of objects in S can be determined by finding the number of objects in each of the parts, adding the numbers obtained:*

$$|S| = |S_1| + |S_2| + \cdots + |S_m|.$$

Example 3.13. Suppose that either a member of the mathematics faculty or a student who is a mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student?

Solution: □

Example 3.14. A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. No project is on more than one list. How many possible projects are there to choose from?

Solution: □

3.3 Subtraction Rule

FACT 3.15 (Subtraction Rule)**.** *If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.*

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Example 3.16. How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

Solution: Let A be the set of all strings of length 8 which starts with 1, B be the set of strings of length 8 which ends in 00. Then, we have $|A| = 2^7$, $|B| = 2^6$ and $|A \cap B| = 2^5$. Therefore, we have $|A \cup B| = 2^7 + 2^6 - 2^5 = 128 + 64 - 32 = 160$. □

Example 3.17. A computer company receives 350 applications from college graduates for a job planning a line of new web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?

Solution: Let C be the set of applicants with major in computer science, B be the set of applicants with major in business. Then we have $|C| = 220$. $|B| = 147$ and $|B \cap C| = 51$. Therefore, we have $|B \cup C| = 220 + 147 - 51 = 316$. We also have $|U| = 350$. Therefore, $|(B \cup C)^c| = 350 - 316 = 34$. □

3.4 Division Rule

FACT 3.18 (Division Rule)**.** *There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w, exactly d of the n ways correspond to way w.*

Example 3.19. Suppose that an automated system has been developed that counts the legs of cows in a pasture. Suppose that this system has determined that in a farmer's pasture there are exactly 572 legs. How many cows are there is this pasture, assuming that each cow has four legs and that there are no other animals present?

Solution: □

Example 3.20. How many different ways are there to seat four people around a circular table, where two seatings are considered the same when each person has the same left neighbor and the same right neighbor?

Solution: Fix one of the people. Then the others will be as if they were on a line. Therefore we have 3! ways. □

3.5 Basic Counting

A great many counting problems can be classified as one of the following types:

1. Count the number of *ordered* arrangements or *ordered* selections of objects
   (a) without repeating any object,
   (b) with repetition of objects permitted (but perhaps limited).
2. Count the number of *unordered* arrangements or *unordered* selections of objects
   (a) without repeating any object,
   (b) with repetition of objects permitted (but perhaps limited).

Instead of distinguishing between nonrepetition and repetition of objects, it is sometimes more convenient to distinguish between selections from a set and a multiset. A **multiset** is like a set except that its members need not be distinct. For example, we might have a multiset M with three a's, one b, two e's, and four d's, that is, 10 elements of 4 different types: 3 of type a, 1 of type b, 2 of type e, and 4 of type d. We shall usually indicate a multiset by specifying the number of times different types of elements occur in it. Thus, M shall be denoted by $\{3 \cdot a, 1 \cdot b, 2 \cdot e, 4 \cdot d\}$. The numbers 3,1,2, and 4 are the repetition numbers of the multiset M. A set is a multiset that has all repetition numbers equal to 1. To include the listed case (b) when there is no limit on the number of times an object of each type can occur (except for that imposed by the size of the arrangement), we allow infinite repetition numbers.

Thus, a multiset in which a and c each have an infinite repetition number and band d have repetition numbers 2 and 4, respectively, is denoted by $\{\infty \cdot a, 2 \cdot b, \infty \cdot c, 4 \cdot d\}$. Arrangements or selections in (1) in which order is taken into consideration are generally called **permutations**, whereas arrangements or selections in (2) in which order is irrelevant are generally called **combinations**.

Example 3.21. How many odd numbers between 1000 and 9999 have distinct digits?

Solution: □

Example 3.22. How many integers between 0 and 10,000 have only one digit equal to 5?

Solution: □

Example 3.23. How many different five-digit numbers can be constructed out of the digits 1, 1, 1, 3, 8?

Solution: □

Example 3.24. How many functions are there from the set $\{1, 2, \ldots, n\}$, where n is a positive integer, to the set $\{0, 1\}$.

1. that are one-to-one?
2. that assign 0 to both 1 and n?
3. that assign 1 to exactly one of the positive integers less than n?

Solution: 1. If $n \leq 2$ we have two 1-1 functions. If $n > 2$ there is none.

2. The values of 1 and n are fixed. Therefore we have 2^{n-2} such functions.
3. n has two choices. The rest is determined by the place of unique 1. Therefore we have $2(n-1)$ such functions.

□

Example 3.25. Show that a natural number $n \geq 1$ has an odd number of divisors (including 1 and itself) if and only if $\sqrt{n}$ is an integer.

Solution: Write $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$. The hypothesis says that $\prod(a_i + 1)$ is odd. Therefore, all of a_i are even. Then $\prod p_i^{a_i/2} = \sqrt{n}$ which is an integer. The converse is similar to the proof that $\sqrt{2}$ is irrational. (Why?) □

Recall that there are $\frac{n!}{(n-k)!}$ 1-1 functions $f : \{1, 2, ..., k\} \to S$ where S is a set with n elements. This means that there are $P(n, k) = \frac{n!}{(n-k)!}$ ordered stings of elements of S of length k. These are called k-**permutations** of S. Any k-element subset of S has $k!$ different orderings (permutations). Therefore, we have

$$\frac{\frac{n!}{(n-k)!}}{k!}$$

k-element subsets of S. They are called the k-**combinations** of S. Hence, we have

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$$

k-combinations of S.

Example 3.26. What is the number of ways to order the 26 letters of the alphabet so that no two of the vowels a, e, i, o, and u occur consecutively?

Solution: There are 21 consonants, and so 21! permutations of the consonants. Since we cannot have two consecutive vowels in our final arrangement, the vowels must be in 5 of the 22 spaces before, between, and after the consonants. Our second task is to put the vowels in these places. There are 22 places for the a, then 21 for the e, 20 for the i, 19 for the 0, and 18 for the u. That is, the second task can be accomplished in $P(22, 5) = \frac{22!}{17!}$ ways. By the multiplication principle, we determine that the number of ordered ar- rangements of the letters of the alphabet with no two vowels consecutive is $21! \times \frac{22!}{17!}$. □

Example 3.27. How many seven-digit numbers are there such that the digits are distinct integers taken from $\{1, 2, ..., 9\}$ and such that the digits 5 and 6 do not appear consecutively in either order?

Solution: Let T be the set of 7-digit numbers formed by using distinct digits from $\{1, 2, 3, ..., 9\}$. Then $|T| = P(9, 7) = \frac{9!}{7!} = 181,440$. Let S be the elements of T in which 5 and 6 do not occur consequently. Then $T - S$ consists of numbers where 5 and 6 occur consequently. Then

$$|S^c| = 2 \times 6 \times P(7, 5) = 30,240.$$

Therefore,

$$|S| = 181,440 - 30,240 = 151,200.$$

□

Ordinary permutations are orderings on the line. If we consider orderings on a circle, we sepak of circular permutations.

THEOREM 3.28 (Circular r-permutations)**.** *The number of circular r-permutations of a set of n elements is given by*

$$\frac{P(n, r)}{r} = \frac{n!}{r \cdot (n - r)!}.$$

COROLLARY 3.29. The number of circular n-permutaions is then

$$\frac{P(n, n)}{n} = \frac{n!}{n \cdot 0!} = (n - 1)!$$

Example 3.30. Ten people, including two who do not wish to sit next to one another, are to be seated at a round table. How many circular seating arrangements are there?

Solution: First seat P_1 at the "head" of the table. Then P_2 cannot be on either side of P_1. There are 8 choices for the person on P_1's left, 7 choices for the person on P_1's right, and the remaining seats can be filled in 7! ways. Thus, the number of seating arrangements in which P_1 and P_2 are not together is

$$8 \times 7 \times 7! = 7 \times 8!.$$

□

Example 3.31. Twenty-five points are chosen in the plane so that no three of them are collinear. How many straight lines do they determine? How many triangles do they determine?

Solution: The number of lines is $\binom{25}{2}$.
There are $\binom{25}{3}$ triangles. □

Example 3.32. How many eight-letter words can be constructed by using the 26 letters of the alphabet if each word contains three, four, or five vowels? It is understood that there is no restriction on the number of times a letter can be used in a word.

Solution:

$$N_3 = \binom{8}{3} 5^3 21^5.$$

$$N_4 = \binom{8}{4} 5^4 21^4.$$

$$N_5 = \binom{8}{5} 5^5 21^3.$$

Hence, $N = N_1 + N_2 + N_3$. □

3.6 Binomial Formula

Let us consider the product $(x + y)^n$. Each term will be of the form $x^p y^q$ where $p + q = n$. When we calculate each term, we can consider the set of words of length n written with the letters x and y. We can write a n-string with k times xs in $\binom{n}{k}$ many ways. Therefore, when we multiply $(x + y)^n = (x + y)(x + y) \cdots (x + y)$, we have $\binom{n}{k}$ many $x^k y^{n-k}$, for each k. Therefore, we have

$$(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

COROLLARY 3.33. We put $x = y = 1$. Then we have

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Therefore, a set with n elements has 2^n subsets. This is the third proof we have for this very important result.

COROLLARY 3.34. Now put $x = 1$ and $y = -1$. Then we have

$$\sum_{k=0}^{n} \binom{n}{k}(-1)^k = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots (-1)^n \binom{n}{n} = 0.$$

Let S be a set with n elements. Let us call a subset $A \subseteq S$ even if it has an even number of elements. Similarly odd subsets are defined. This corollary says that the number of even subsets of S and the number of odd subsets of S are equal.

There are many properties of these coefficients: For example, one can show very easily that

$$\binom{n}{k} = \binom{n}{n-k}.$$

THEOREM 3.35 (Pascal's Identity)**.** *Let n and k be positive integers with $k \leq n$. Then we have*

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

Proof. Let S be a set havein $n+1$ elements. Write $S = \{x_1, x_2, ..., x_n, y\}$. Let $\mathcal{A} = \{A \subseteq S : |A| = k\}$, $\mathcal{B} = \{A \in \mathcal{A} : y \in A\}$ and $\mathcal{C} = \{A \in \mathcal{A} : y \notin A\}$. Then $\mathcal{A} = \mathcal{B} \cup \mathcal{C}$. There is a 1-1 correspondence between $\mathcal{B}$ and the set of subsets of $S \setminus \{y\}$ with $k-1$ elements. Therefore $\mathcal{B} = \binom{n}{k-1}$ and $\mathcal{C} = \binom{n}{k}$. Clearly $\mathcal{B}$ and $\mathcal{C}$ are disjoint. Therefore we have

$$\binom{n+1}{k} = |\mathcal{A}| = \binom{n}{k-1} + \binom{n}{k}.$$

□

Induction Proof of the Binomial Formula. It is enough to prove the following identity

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

The binomial formula is clearly true for $n = 1$. Suppose that the formula holds for

some $n \in \mathbb{N}$. We have

$$\begin{aligned}
(1+x)^{n+1} &= (1+x)^n(1+x) \\
&= \left(\sum_{k=0}^{n} \binom{n}{k} x^n\right) \cdot (1+x) \\
&= \sum_{k=0}^{n} \binom{n}{k} x^n + \sum_{k=0}^{n} \binom{n}{k} x^{n+1} \\
&= 1 + \left[\binom{n}{1} + \binom{n}{0}\right] x + ... + \left[\binom{n}{n} + \binom{n}{n-1}\right] x^n + x^{n+1} \\
&= 1 + \binom{n+1}{1} x + \binom{n+1}{2} x^2 + ... + \binom{n+1}{n} x^n + x^{n+1} \\
&= \sum_{n=0}^{n+1} \binom{n+1}{k} x^k.
\end{aligned}$$

□

Example 3.36. Consider $X = \{A \subset [n] : |A| = 2\}$.

For each $i \leq n$ let $X_i = \{A \in X : \max(A) = i\}$.

Then we have $|X_1| = 0$, $|X_2| = 1$, $|X_3| = 2$,...,$|X_i| = i - 1$,...,$|X_n| = n - 1$.

We also have $X = X_1 \cup X_2 \cup X_3 \cup ... \cup X_n$ (disjoint). Therefore, we have $|X| = \sum |X_i|$, hence

$$0 + 1 + 2 + \cdots + (n-1) = \binom{n}{2} = \frac{n.(n-1)}{2}.$$

3.7 Permutations of Multisets

If S is a multiset, an r-permutation of S is an ordered arrangement of r of the objects of S. If the total number of objects of S is n (counting repetitions), then an n-permutation of S will also be called a permutation of S.

THEOREM 3.37. *Let S .be a multiset with objects of k different types, where each object has an infinite repetition number. Then the number of r-permutations of S is k^r.*

Example 3.38. What is the number of ternary numerals with at most four digits?

THEOREM 3.39. *Let S be a multiset with objects of k different types with finite repetition numbers $n_1, n_2, ..., n_k$, respectively. Let the size of S be $n = n_1+n_2+...+n_k$. Then the number of permutations of S equals*

$$\frac{n!}{n_1! \cdot n_2! \cdots n_k!}.$$

Idea of proof:

$$\binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-\ldots n_{k-1}}{n_k}=\frac{n!}{n_1!n_2!\ldots n_k!}.$$

□

Example 3.40. The number of permutations of the letters in the word MISSISSIPPI is

$$\frac{11!}{1!4!4!2!}.$$

Solution: Here our multiset is $\{1\cdot M, 4\cdot I, 4\cdot S, 2\cdot P\}$. The rest follows. □

Example 3.41. Let $S=\{a,b,c,d\}$. Partition S into 2 sets of equal size. Then we have 3 such partitions:

$$\{a,b\},\{c,d\};\quad \{a,c\},\{b,d\},\quad \{a,d\},\{b,c\}.$$

Now suppose that the parts are labeled by e.g colors red and blue. Then we have 6 of them:....For example, for the partition $\{a,b\},\{c,d\}$, we have

$$\{a,b\},\{c,d\}$$

$$\{a,b\},\{c,d\}$$

This way we have 6 labeled partitions.

THEOREM 3.42. *Let n be a positive integer and let $n_1,n_2,\ldots,n_k$ be positive integers with $n=n_1+n_2+\ldots+n_k$ The number of ways to partition a set of n objects into k labeled boxes in which Box 1 contains n_1 objects, Box 2 contains n_2 objects, ... , Box k contains n_k objects equals*

$$\frac{n!}{n_1!\cdot n_2!\cdots n_k!}.$$

If the boxes are not labeled and $n_i=n_j$ for every i,j, then it is equal to

$$\frac{n!}{k!n_1!n_2!\ldots n_k!}.$$

Example 3.43. Let $S=\{3\cdot a, 2\cdot b, 4\cdot c\}$. Find the number of 8-permutations of S.

Solution:

1. $S_1=\{2\cdot a,2\cdot b,4\cdot c\}$. Then we have $N_1=\frac{8!}{2!2!4!}=420$
2. $S_2=\{3\cdot a,1\cdot b,4\cdot c\}$. Then we have $N_2=\frac{8!}{3!1!4!}=280$.
3. $S_3=\{3\cdot a,2\cdot b,3\cdot c\}$. Then we have $N_3=\frac{8!}{3!2!3!}=560$.

The answer is $N=N_1+N_2+N_3$. □

3.8 Pigenhole Principle

Pigeonhole Principle: Simple Form

THEOREM 3.44. *If $n+1$ objects are distrbuted in n boxes, then at least one of the boxes contains two or more objects.*

One can prove this statement by contradiction!

Example 3.45.
1. Among any 13 people there are 2 who have their birthdays in the same month.
2. There are n married couples. This way we have in total $2n$ people. Among any $n+1$ of then, there is a married couple.

Some variants:

1. If n objects are put into n boxes and if no box is empty, then each box contains exactly one object.
2. If n objects are put into n boxes and no box gets more than one object, then every box has an object in it.

Another Formulation:
Suppose that X and Y are two finite sets and let $f : X \to Y$ be a function.

1. If $|Y| < |X|$, then f can not be 1-1.
2. If $|X| = |Y|$ and f is onto, then it is also 1-1.
3. If $|X| = |Y|$ and f is 1-1, then it is onto.

Example 3.46. From the integers 1,2, ... ,200, we choose 101 integers. Show that, among the integers chosen, there are two such that one of them is divisible by the other.

Idea. Every integer is of the form $2^k a$ where k is a nonnegative integer and a is an odd integer. □

THEOREM 3.47. *If $mn+1$ pigeons are placed into n pigeonholes, then there is a pigeonhole with at least $m+1$ pigeons.*

THEOREM 3.48. *Let $m_1, m_2, ..., m_n$ be positive integers. If $m_1 + m_2 + ... + m_n - n + 1$ pigeons are placed into n pigeonholes, then for some $1 \le i \le n$ such that ith pigeonhole contains at least m_i pigeons.*

Again, one can prove this statemnt by contradiction!.

Example 3.49. There is a nonzero multiple of 6 whose digits are all zeros or ones.

Idea. Consider the any 6 different numbers consisting of only ones. Write $a_1 < a_2, ... < a_6$. If one of them is divisible by 6, we are done. Suppose not. Let r_1 be the remainder when we divide a_1 by 6, r_2 be the remaninder when we divide a_2 by 6,... Then $r_1, r_2, ..., r_6 \in \{1, 2, 3, 4, 5\}$. By the pigeonhole principle there are $1 \leq i < j \leq 6$ such that $r_i = r_j$. Now it is easy to see that $a_j - a_i$ has digits equal to either 0 or 1, and $a_j - a_i$ is divisible by 6. □

Example 3.50. Let S be a set consisting of $n \geq 2$ integers. Then there is a subset A of S such that the sum of elements of A is a multiple of n.

Solution: Let $S_1 = \{a_1\}$, $S_2 = \{a_1, a_2\}$,...,$S_n = S$. Let $s_k = a_1 + a_2, .. + a_k$ be the sum of elements in S_k. If there is some k such that $n|s_k$, we have nothing to prove. If not, by the pigeon principle, there are $k < l$ such that s_k and s_l have the same remainders when divided by n. (Look at the previous example!) This means that $s_l - s_k$ is divisible by n. But $s_l - s_k$ is nothing but the sum of elements in $A - S_l - S_k \subset S$. □

3.9 More on Binomial Coefficients

THEOREM 3.51 (Vandermonde's Identity)**.** *m, n, r are positive integers with $r \leq m$ and $r \leq n$. Then we have*

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{r-k}\binom{n}{k}.$$

Proof. □

COROLLARY 3.52. Let n be a positive integer. Then we have

$$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2.$$

3.10 Exercises

1. Show that if n is a positive integer, then
$$\binom{n}{0} < \binom{n}{1} < \binom{n}{2} < \cdots < \binom{n}{\lfloor \frac{n}{2} \rfloor} = \binom{n}{\lceil \frac{n}{2} \rceil} > \cdots > \binom{n}{n-1} > \binom{n}{0}.$$

2. Show that $\binom{n}{k} \leq 2^n$ whenever n is a positive integer and $0 \leq k \leq n$.

3. (a) If n is an integer with $n > 1$, show that $\binom{n}{\lceil \frac{n}{2} \rceil} \geq \frac{2^n}{n}$.

   (b) If n is a positive integer, show that $\binom{2n}{n} \geq \frac{4^n}{2n}$.

4. If $n, k \in \mathbb{N}$ with $1 \leq k \leq n$, we have
$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

5. Show that $\sum_{k=1}^{n} k\binom{n}{k} = n \cdot 2^{n-1}$.

6. Prove that
$$\binom{n}{r}\binom{r}{k} = \binom{n}{k}\binom{n-k}{r-k}$$
whenever $k, n, r \in \mathbb{N}$ with $k \leq r \leq n$.

7. (Hockeystick Identity) Show that
$$\sum_{k=0}^{r} \binom{n+k}{k} = \binom{n+r+1}{r}$$
whenever n and r are positive integers.

8. One can go either 1 unit in the $+x$ direction or 1 unit in the $+y$ direction at each step. Let $A = (0, 0)$, $B = (m, n)$. If $n(A, B)$ denotes the number of paths from A to B, we have
$$n(A, B) = \binom{n+m}{m}.$$

9. Prove that
$$\sum_{k=0}^{n} k^2\binom{n}{k} = n(n+1)2^{n-2}$$
if n is a positive integer.

3.11 Inclusion-Exclusion Principle

We know
$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Similarly, we have

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

More genrally, let $A_1, A_2, ..., A_n$ be sets. Let S_1 be the sum of $|A_i|$'s. Let S_2 be the sum of the $|A_i \cap A_j|$s where i and j are different. Let S_3 be the some of all possible $|A_i \cap A_j \cap A_k|$'s,... Then we have

$$|A_1 \cup A_2 \cup \cdots \cup A_n| = S_1 - S_2 + S_3 - \cdots + (-1)^{n-1} S_n.$$

Example 3.53. How many integers in $\{1, 2, 3, ..., 500\}$ are not divisible by 4, 6 or 15?

3.12 Tree Diagrams

Example 3.54. How many bit strings of length four do not have two consecutive 1s?

Example 3.55. A playoff between two teams consists of at most five games. The first team that wins three games wins the playoff. In how many different ways can the playoff occur?

Example 3.56. Suppose that "I Love New Jersey" T-shirts come in five different sizes: S, M, L, XL, and XXL. Further suppose that each size comes in four colors, white, red, green, and black, except for XL, which comes only in red, green, and black, and XXL, which comes only in green and black. How many different shirts does a souvenir shop have to stock to have at least one of each available size and color of the T-shirt?

3.13 Mixed Examples

Example 3.57. How many positive integers between 1000 and 9999 inclusive
a) are divisible by 9?
b) are even?
c) have distinct digits?
d) are not divisible by 3?
e) are divisible by 5 or 7?
f) are not divisible by either 5 or 7?
g) are divisible by 5 but not by 7?
h) are divisible by 5 and 7?

Example 3.58. How many bit strings of length 10 either begin with three 0s or end with two 0s?

Example 3.59. How many bit strings of length seven either begin with two 0s or end with three 1s?

Example 3.60. How many bit strings of length eight contain either three consecutive 0s or four consecutive 1s?

Example 3.61. How many solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 + x_5 = 21,$$

where x_i , $i = 1, 2, 3, 4, 5$, is a nonnegative integer such that
a) $x_1 \geq 1$?
b) $x_i \geq 2$ for $i = 1, 2, 3, 4, 5$?
c) $0 \leq x_1 \leq 10$?
d) $0 \leq x_1 \leq 3$, $1 \leq x_2 < 4$, and $x_3 \geq 15$?